Java Assignment Tracker

Project Design Document

GitHub Repository: https://github.com/Prachet4316/Java-Assignment-Tracker

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1.1 Problem Statement

I personally often struggle to keep track of assignments from different courses and their individual deadlines, which I think is a problem shared by a lot of students. When the assignments are given to us through different sources, maybe verbally or on different platforms, collecting all of it into an organized tracker app would make our quality of life better

1.2 Proposed Solution

I am going to build a simple Assignment Tracker. The aim is for students to be able to enter an assignment with its details. The application will display incomplete assignments sorted by the deadline days. Students will be allowed to mark assignments as completed, and for now, the assignments can be removed once they pass their deadlines. Assignments should also be saved so they remain available when reopened

2. Eisenhower Matrix

DO - MVP	Schedule
1. Add assignments with course name, assignment name, and due date. 2. Display incomplete assignments in the application. 3. Sort assignments by closest deadline. 4. Mark assignments as completed. 5. Save assignments between sessions. 6. Automatically remove expired assignments.	1. Add assignment priority labels. 2. Add search and filtering by course. 3. Display weekly and monthly assignment statistics. 4. Add reminders for upcoming deadlines.
Delegate	**Delete**
1. Use Java date/time libraries to handle deadlines. 2. Use Java file-handling utilities to save and load assignments. 3. Use Java Swing components to create the graphical interface.	1.Online account registration and login. 2. Cloud synchronization across multiple devices. 3. Social features for sharing assignments with friends. 4. AI-generated study schedules.

3. Technology Stack

Java - Main programming language

Java Swing - Creating the graphical user interface

ArrayList - Storing and managing assignments while the application runs

LocalDate - Handling assignment due dates

File I/O - Reading and writing assignment data

CSV - Persisting assignment information locally

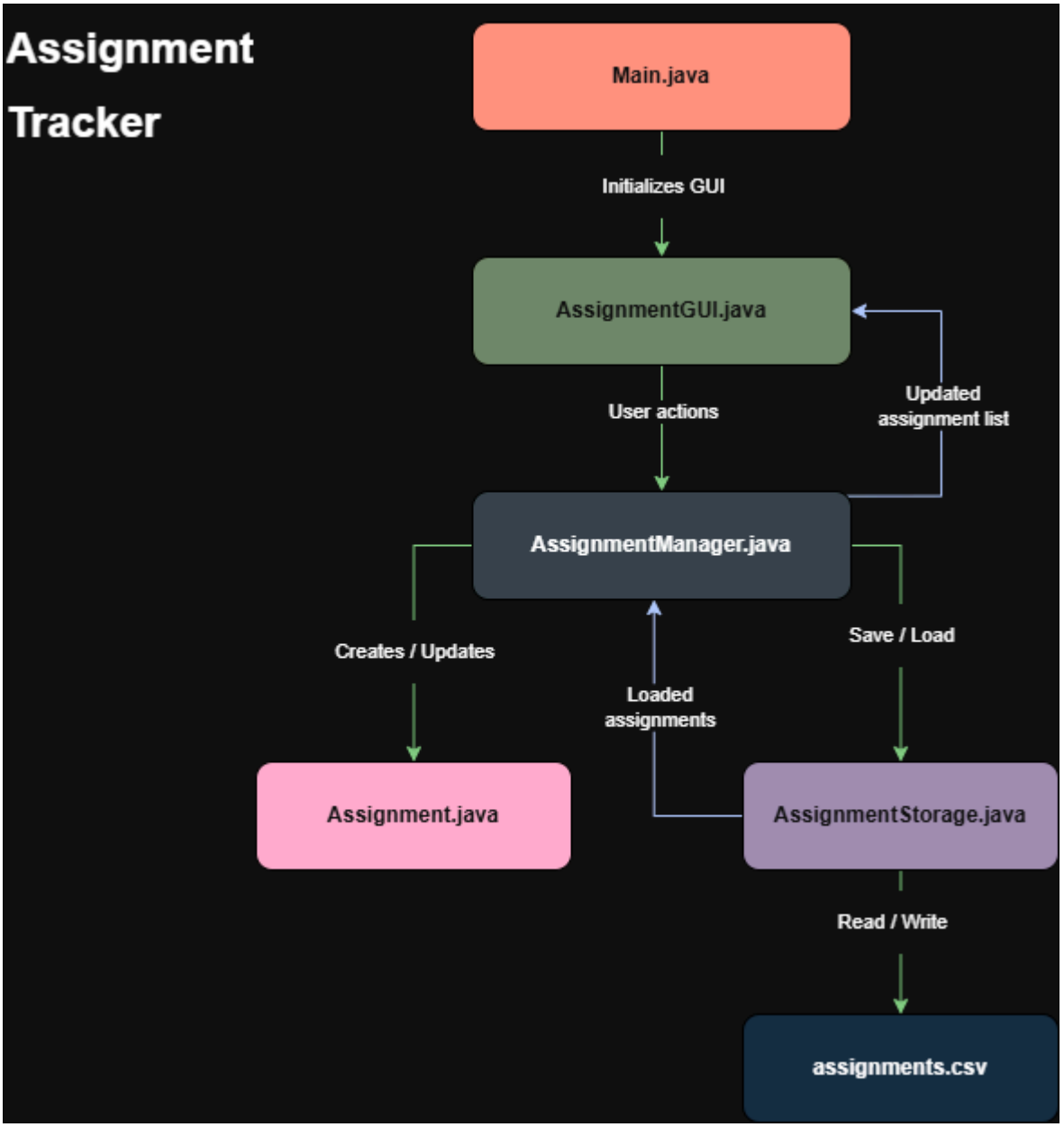
Git and GitHub - Version control and project management

4. Development Plan

The project will be developed in the following milestones:

1. Set up the GitHub repository and Java project structure.
2. Create the Assignment class with the required fields, constructor, and methods.
3. Implement CSV storage to save and load assignments.
4. Build the basic graphical interface using Java Swing.
5. Implement the ability to add new assignments.
6. Implement the ability to mark assignments as completed.
7. Implement deadline sorting and automatic removal of expired assignments.
8. Test the application, fix issues, and polish the interface and README.

5. System Diagram



6. GitHub Repository

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